

MAT1575, Classwork6, Fall2026

1.6 Integrals Involving Exponential and Logarithmic Functions (p.84-87, 88-91)

1. Integrals of Exponential Functions.

Exponential functions can be integrated using the following formulas:

$$\int e^x dx = e^x + C \text{ and } \int a^x dx = \frac{a^x}{\ln a} + C$$

2. Show $\int a^x dx = \frac{a^x}{\ln a} + C$

since

u-sub, let $u = x \ln a$, then

$$\int a^x dx = \int e^{x \ln a} dx \stackrel{u}{=} \frac{1}{\ln a} e^{x \ln a} + C = \frac{a^x}{\ln a} + C$$

3. Find the antiderivative of the exponential function e^{-x} .

$$\int e^{-x} dx = -e^{-x} + C$$

($u = -x \rightarrow -du = dx$
 $du = -dx \rightarrow \int e^u du = -e^u + C = -e^{-x} + C$)

4. Find the antiderivative of the exponential function $e^x \sqrt{1+e^x}$.

$$\int e^x \sqrt{1+e^x} dx = \int \sqrt{u} du = \int u^{\frac{1}{2}} du$$

$u = 1+e^x$
 $du = e^x dx$

$$= \frac{2}{3} u^{\frac{3}{2}} + C = \frac{2}{3} (e^x + 1)^{\frac{3}{2}} + C = \frac{2}{3} (\sqrt{e^x + 1})^3 + C$$

5. Evaluating a Definite Integral Using Substitution

(a) $\int_1^2 \frac{e^{\frac{1}{x}}}{x^2} dx$ $u = \frac{1}{x}, du = -\frac{1}{x^2} dx$
 $-du = \frac{1}{x^2} dx$

$$= - \int_{\frac{1}{2}}^1 e^u du$$

$$= \int_{\frac{1}{2}}^1 e^u du = [e^u]_{\frac{1}{2}}^1 = e^1 - e^{\frac{1}{2}} = e - \sqrt{e}$$

(b) $\int_1^2 \frac{1}{x^3} e^{4x^{-2}} dx$ $u = 4x^{-2}, du = -8x^{-3} dx$
 $= \frac{4}{x^2} = -\frac{8}{x^3} dx \Rightarrow \frac{du}{-8} = \frac{1}{x^3} dx$

$$= \int_{u(2)}^{u(1)} e^u \cdot \frac{du}{-8}$$

$$= \int_4^1 e^u \frac{du}{-8}$$

$$= -\frac{1}{8} \int_4^1 e^u du$$

$$= \frac{1}{8} \int_1^4 e^u du$$

$$= \frac{1}{8} [e^u]_1^4 = \frac{1}{8} [e^4 - e^1]$$

6. Integration Formulas Involving Logarithmic Functions.

The following formulas can be used to evaluate integrals involving logarithmic functions.

$$\int \frac{1}{x} dx = \ln |x| + C$$

$$\int \ln x dx = x \ln x - x + C \text{ and } \int \log_a x dx = \frac{x \ln x - x}{\ln a} + C$$

Note: $\log_a x = \frac{\ln x}{\ln a}$ implies $\int \log_a x dx = \int \frac{\ln x}{\ln a} dx = \frac{1}{\ln a} \int \ln x dx$

$$= \frac{1}{\ln a} (x \ln x - x) + C$$

7. Find the antiderivative of the function $\frac{3}{x-10}$. $u=x-10, du=dx$

$$\int \frac{3}{x-10} dx = 3 \int \frac{du}{u} = 3 \ln|u| + C = 3 \ln|x-10| + C$$

8. Find the antiderivative of the function $\frac{2x^3+3x}{x^4+3x^2}$.

$$\int \frac{2x^3+3x}{x^4+3x^2} dx \quad u=x^2+3x^2, du=(4x^3+6x)dx \Rightarrow \frac{du}{2} = (2x^3+3x)dx$$

$$= \int \frac{1}{u} \frac{du}{2} = \frac{1}{2} \int \frac{du}{u} = \frac{1}{2} \ln|u| + C = \frac{1}{2} \ln|x^4+3x^2| + C$$

9. Find the definite integral of $\int_0^{\pi/2} \frac{\sin \theta}{1+\cos \theta} d\theta$

$$u=1+\cos(\theta) \quad du=-\sin(\theta)d\theta$$

$$-\int du = \sin(\theta)d\theta$$

$$= \int_{u(0)}^{u(\pi/2)} \frac{-du}{u} = -\int_2^1 \frac{du}{u} = \int_1^2 \frac{du}{u}$$

$$= [\ln|u|]_1^2 = \ln(2) - \ln(1) = \ln 2$$

10. Find the antiderivative of the function $\sec \theta$. Hint: Let $u = \sec \theta + \tan \theta$.

$$\int \sec \theta d\theta = \int \sec \theta \frac{(\sec \theta + \tan \theta)}{(\sec \theta + \tan \theta)} d\theta \quad du = (\sec \theta \tan \theta + \sec^2 \theta) d\theta = \sec \theta (\tan \theta + \sec \theta) d\theta$$

$$= \int \frac{du}{u} = \ln|u| + C = \ln|\sec \theta + \tan \theta| + C$$

11. Find the antiderivative of the function $\csc \theta$. Hint: Let $u = \csc \theta - \cot \theta$. *leave as an exercise.*

$$\int \csc \theta \frac{(\csc \theta - \cot \theta)}{(\csc \theta - \cot \theta)} d\theta \rightarrow du = -\csc \theta \cot \theta + \csc^2 \theta$$

$$= \int \frac{du}{u} = \ln|u| + C = \ln|\csc \theta - \cot \theta| + C$$

12. Find the antiderivative of the function $(\tan \theta) \ln(\cos \theta)$.

Let $u = \cos(\theta)$, we have $du = -\sin(\theta)d\theta$. Then $-du = \sin \theta d\theta$

$$\int (\tan \theta) \ln(\cos \theta) d\theta = \int \frac{\sin \theta}{\cos \theta} \ln(\cos \theta) d\theta = \int \frac{\ln(u)}{u} \cdot (-du) = -\int \frac{\ln(u)}{u} du$$

13. Preparation for Integration By Parts (section 3.1):

The acronym **LIATE** stands for:

L is Logarithmic Functions: $\ln(x), \log_2(x), \log_{10}(x)$

I is Inverse Trigonometric Functions: $\sin^{-1}(x), \cos^{-1}(x),$

A is Algebraic Functions: polynomial: x^3+x^2+1

T is Trigonometric Functions: $\sin(\theta), \cos(\theta), \tan(\theta)$

E is Exponential Functions: $e^x, 2^x, 5^x, e^{3x}, e^{-x}$

$$V = \ln(u) \quad dV = \frac{du}{u}$$

$$= -\int V dV$$

$$= -\frac{V^2}{2} + C$$

$$= -\frac{(\ln(u))^2}{2} + C = -\frac{(\ln|\cos(\theta)|)^2}{2} + C$$